Reg. No. | | | | | | | | | | | [18AS741]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

SPACE DYNAMICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	What do you understand by the moment of a vector? Explain it with a neat sketch.	**06**
	b.	What do you understand by inertial and non-inertial frames? Derive the expression for the inertial derivative of a body-fixed vector.	**08**
2.	**a.**	Write down Kepler's laws of planetary motion. Using Kepler's third law, how to find the ratio of the mass of the Sun to the mass of the Earth?	**07**
	b.	What do you understand by central force? Prove that the angular momentum is constant when two bodies are moving under the influence of mutual gravitational attraction.	**07**
3.	**a.**	What is trajectory transfer? Describe the types of the trajectory transfer mechanism based on the center of force.	**06**
	b.	Derive the expression for the minimum eccentricity of the transfer orbit.	**08**
		OR	
4.	**a.**	What do you understand by coplanar maneuver? Write down the orbital parameters which will be affected during the coplanar maneuver.	**04**
	b.	In order to change the orbit of a satellite, calculate the change in velocity required at the apogee and perigee location of the minimum eccentricity transfer orbit.	**10**
5.	**a.**	Derive the expression for the moment equations (gyroscope equations) in terms of gimbal rates.	**07**
	b.	Describe the motion of a spinning top with a neat sketch.	**07**
		OR	
6.	**a.**	For a torque-free motion of axisymmetric bodies, show that angular momentum vector, angular velocity vector, and the unit vector along the spin axis lie in one plane at all times.	**08**
	b.	Consider a satellite is spinning about an axis of minimum inertia. Comment on the stability of the satellite motion using appropriate mathematical equations.	**06**
7.	**a.**	Write down the general equations (moment and force equations) in body coordinates where the body axes are not coinciding with principal axes.	**07**
	b.	Describe the motion of a spinning missile with the misalignment of the thrust line. For such situations, write down Euler's equation for the missile.	**07**
		OR	
8.	**a.**	Derive the equation for a rocket in rectilinear motion.	**06**
	b.	What do you understand by despinning of satellites? Describe the attitude drift of space vehicles.	**08**